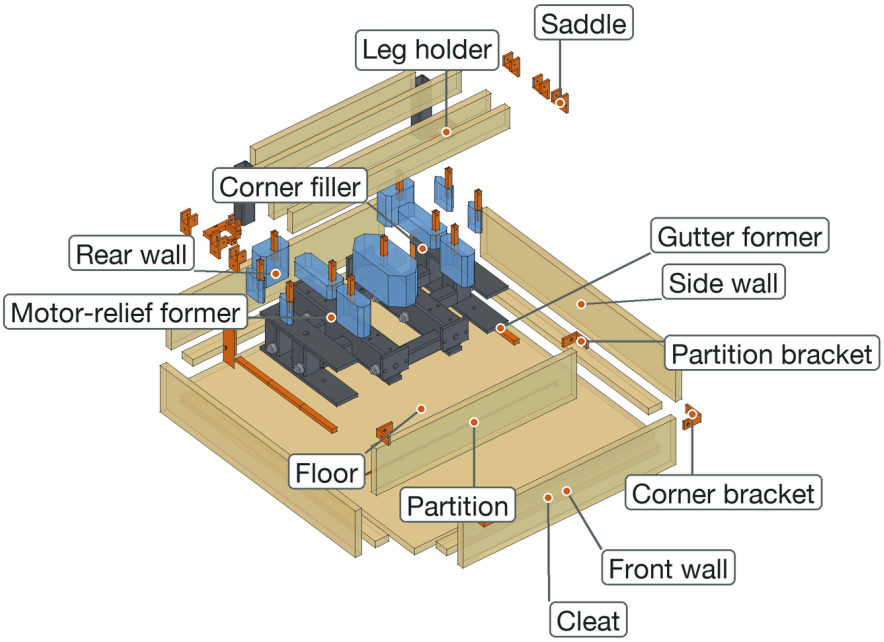
Base mold

DIY CNC epoxy-granite mill · October 2026

How to make the base mold. Drilling templates: `templates/base-templates.pdf`. The complete guide (all the molds and the sheet layout for the boards) is `molds-complete.pdf`.

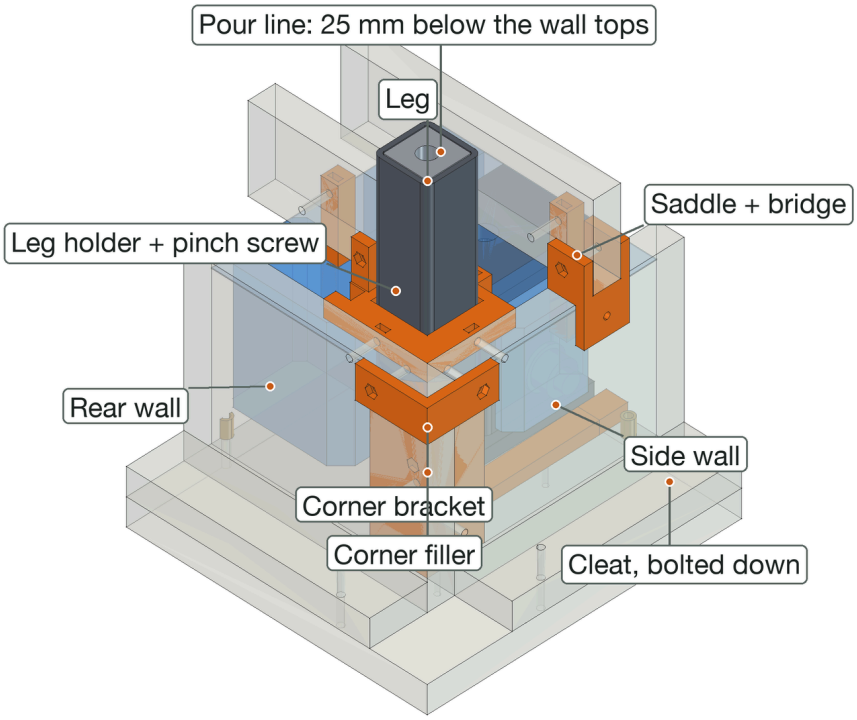


In short: melamine boards bolt together with printed brackets. Every printed piece bolts or pins to its board from outside. The walls stand 25 mm above the pour line, so no bolt goes through a face the mix touches. Nothing is glued, and it all comes apart for the next casting.

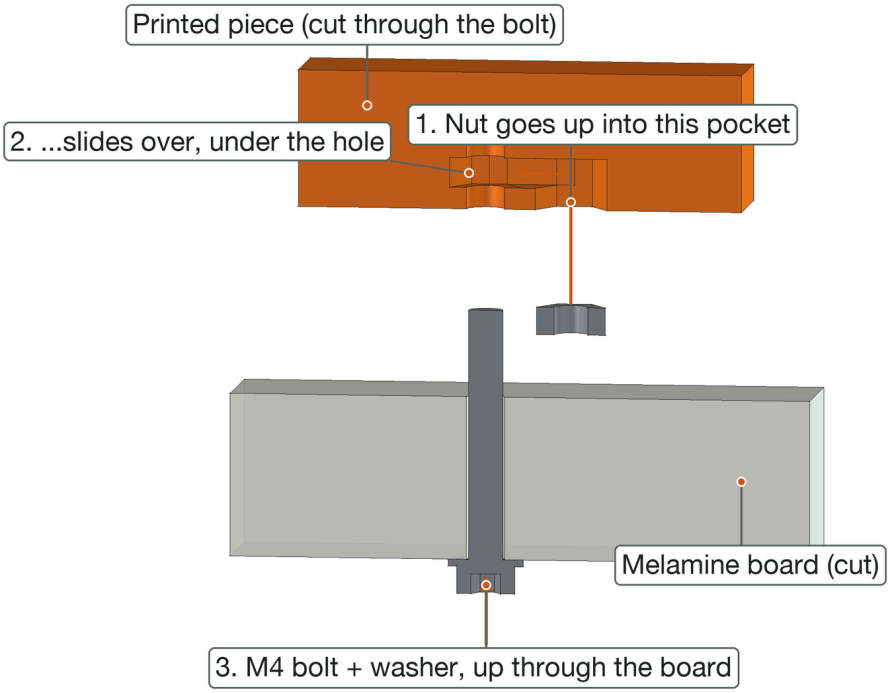
Order of work: print → cut the boards → mark them with the templates and drill → cut the foam → wax → assemble → pour → take apart.

1. How it works

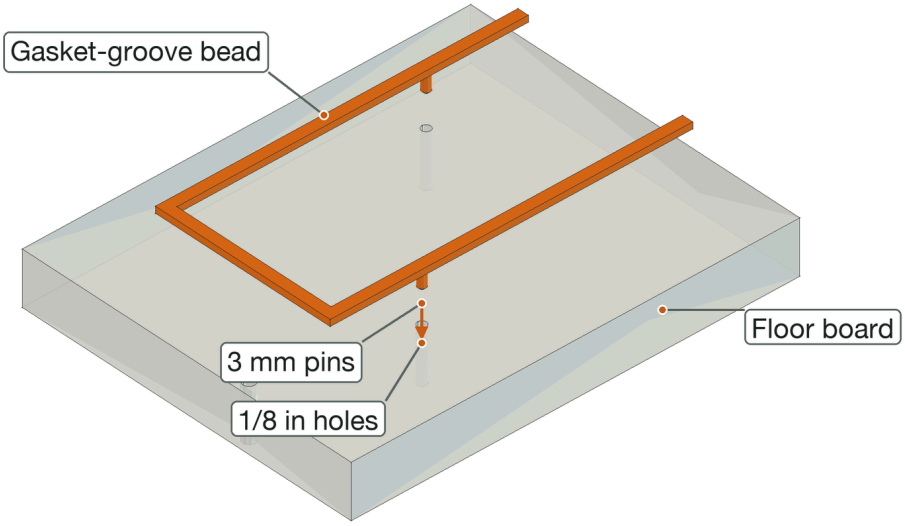
- **Walls** stand on the floor against bolted-down **cleats**; **corner brackets** square the tops. All the bolts go through the top 25 mm, above the pour line.



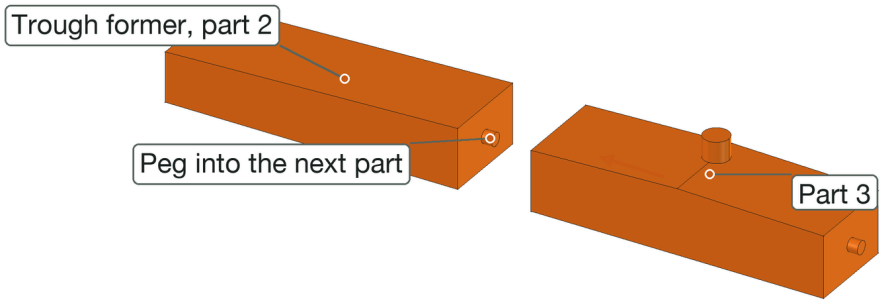
- **Printed pieces** have a nut built in: it goes into a pocket on the board side and slides under the hole; an M4 bolt comes through the board from outside.



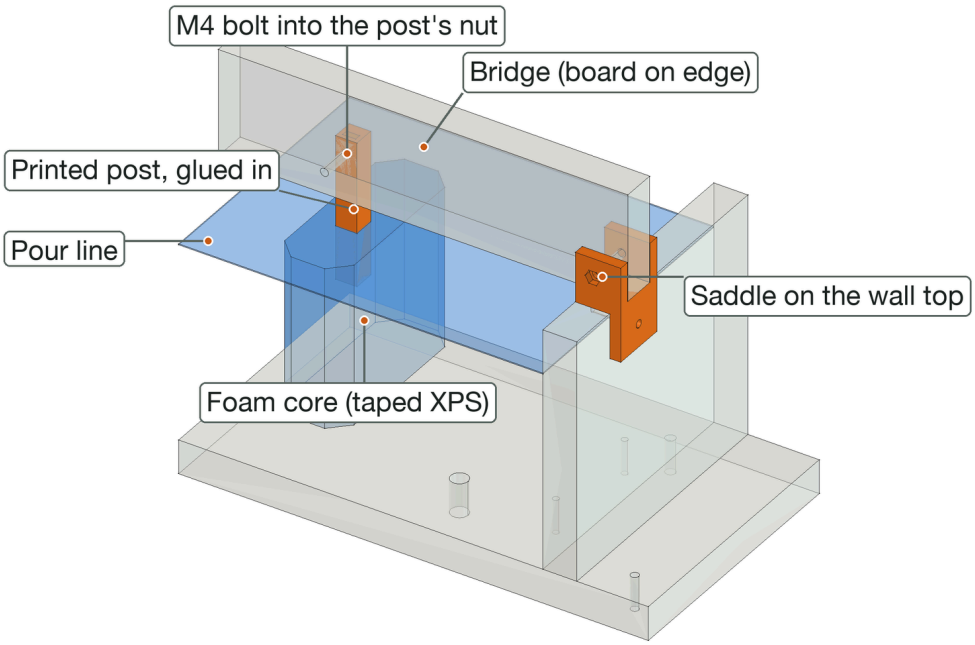
- **Thin pieces** (gasket-groove beads, motor relief) have 3 mm pins that push into 1/8 in holes.



- **Long pieces** are printed in parts that peg together; each part bolts down on its own.



- **Cores** are taped foam blocks with a printed post that bolts to a bridge across the mold.



- **Fits** are 0.15 mm a side, and drawn for 19.05 mm board (Home Depot's 3/4 in melamine is a true 0.750 in). Measure yours; if it differs, set `mold_board` in `model/build_cnc.py` and re-run it.

2. What you need

Material	Amount	Notes
Melamine board, 3/4 in	about 1.5 m2	all the molds come out of two 4 x 8 ft sheets
XPS foam board, 2 in	1 sheet, 2 x 8 ft	the cores
PETG	0.64 kg	sliced; everything printed for the machine fits in two 1 kg rolls
Packing tape, double-sided carpet tape	1 roll each	wrapping and holding the foam
Paste wax and PVA	1 each	release
3/8 in quarter-round PVC molding	a few meters	the rounded outside edges

Hardware	Qty
M3 nut	20
M3 washer	20
M3 x 25 bolt	20
M4 nut	113
M4 washer	109
M4 x 20 bolt (leg pinch screw)	4
M4 x 30 bolt	65
M4 x 35 bolt	28

Hardware	Qty
M4 x 45 bolt	16
M5 x 20 screw, greased (holds an insert)	11
M5 x 30 screw, greased (holds a rail bar)	10
M6 x 30 screw, greased (holds a support pad)	6

Add about 10 % spare small bolts, nuts and washers. **Tools:** drill and brad-point bits (sizes in section 5), a 2-1/2 in hole saw, a square, clamps, a 7 mm nut driver, a 3 mm hex key, an awl, a fine saw or hot wire for the foam.

3. Print

71 prints, 0.64 kg, about 57 hours (sliced; settings in the print guide). Print one corner bracket and one saddle first and check them on a board offcut. Press or slide the nuts in before the pieces go on the boards.

File	Qty	g	Time each
`base-box/kit-corner-bracket.stl`	4	33	39m 23s
`base-box/kit-partition-bracket.stl`	2	14	35m 28s
`base-box/kit-saddle.stl`	6	58	46m 3s
`base-box/kit-leg-holder.stl`	4	82	1h 37m 0s
`base-rear/former-gutter-L-part1of3.stl`	1	8	35m 5s
`base-rear/former-gutter-L-part2of3.stl`	1	8	33m 40s
`base-rear/former-gutter-L-part3of3.stl`	1	7	31m 15s
`base-rear/former-gutter-R-part1of3.stl`	1	8	35m 6s
`base-rear/former-gutter-R-part2of3.stl`	1	8	33m 40s
`base-rear/former-gutter-R-part3of3.stl`	1	7	31m 15s
`base-rear/former-motor-relief.stl`	1	12	40m 58s
`base-rear/filler-corner-L.stl`	1	17	1h 44m 24s
`base-rear/filler-corner-R.stl`	1	16	1h 42m 27s
`base-front/former-gutter-L-part1of2.stl`	1	11	49m 23s
`base-front/former-gutter-L-part2of2.stl`	1	10	43m 43s
`base-front/former-gutter-R-part1of2.stl`	1	11	49m 24s
`base-front/former-gutter-R-part2of2.stl`	1	10	43m 43s
`base-front/former-trough-part1of5.stl`	1	20	1h 14m 18s
`base-front/former-trough-part2of5.stl`	1	22	1h 23m 56s
`base-front/former-trough-part3of5.stl`	1	25	1h 33m 33s
`base-front/former-trough-part4of5.stl`	1	23	1h 24m 3s
`base-front/former-trough-part5of5.stl`	1	21	1h 17m 36s
`base-front/filler-corner-L.stl`	1	16	1h 43m 48s

File	Qty	g	Time each
base-front/filler-corner-R.stl	1	17	1h 40m 33s
base-rear/former-split-gasket-groove-part1of4.stl	1	2	9m 50s
base-rear/former-split-gasket-groove-part2of4.stl	1	1	8m 58s
base-rear/former-split-gasket-groove-part3of4.stl	1	1	8m 58s
base-rear/former-split-gasket-groove-part4of4.stl	1	2	9m 53s
base-front/core-center-a-post.stl	1	5	44m 56s
base-front/core-center-b-post.stl	1	5	44m 56s
base-rear/core-center-post.stl	1	5	44m 56s
base-front/core-outboard-L-post.stl	1	5	44m 53s
base-front/core-trough-band-L-post.stl	1	5	44m 54s
base-front/core-draw-rod-L-post.stl	1	5	44m 53s
base-rear/core-foot-b-L-post.stl	1	5	44m 53s
base-rear/core-knee-L-post.stl	1	5	44m 53s
base-rear/core-outer-a-L-post.stl	1	5	44m 53s
base-rear/core-outer-b-L-post.stl	1	5	44m 53s
base-rear/core-rear-L-post.stl	1	5	44m 53s
base-front/core-outboard-R-post.stl	1	5	44m 53s
base-front/core-trough-band-R-post.stl	1	5	44m 54s
base-front/core-draw-rod-R-post.stl	1	5	44m 53s
base-rear/core-foot-b-R-post.stl	1	5	44m 53s
base-rear/core-knee-R-post.stl	1	5	44m 53s
base-rear/core-outer-a-R-post.stl	1	5	44m 53s
base-rear/core-outer-b-R-post.stl	1	5	44m 53s
base-rear/core-rear-R-post.stl	1	5	44m 53s
base-front/plug-draw-pocket-L.stl	1	12	46m 16s
base-front/plug-draw-pocket-R.stl	1	12	46m 16s
base-rear/plug-draw-pocket-L.stl	1	5	21m 23s
base-rear/plug-draw-pocket-R.stl	1	5	21m 23s
tools/kit-pour-gauge.stl	2	9	24m 33s
tools/kit-screed-shoe.stl	2	20	44m 42s
base-front/cast-in-leg-front-l-plug.stl	2	8	24m 14s
base-rear/cast-in-leg-rear-l-plug.stl	2	8	24m 14s

4. Cut the boards

Have the store rip the strips; crosscut at home. Write each board's name on its outside face, and mark the cast face (the melamine face the mix touches).

Board	Qty	Size (mm)	Notes
Base floor	1	718 x 843	both halves are cast on it
Base front wall	1	638 x 135	
Base rear wall	1	638 x 135	
Base side wall, left	1	724 x 135	
Base side wall, right	1	724 x 135	
Base partition	1	600 x 135	rear half only; comes out before the front half
Base partition strut	2	326 x 60	offcut, stands on edge; holds the partition foot (rear half only)
Base cleat, front	1	638 x 40	offcut strip, flat on the floor
Base cleat, rear	1	638 x 40	offcut strip, flat on the floor
Base cleat, left	1	724 x 40	offcut strip, flat on the floor
Base cleat, right	1	724 x 40	offcut strip, flat on the floor
Base bridge y=-230	1	651 x 60	
Base bridge y=-130	1	651 x 60	
Base bridge y=-60	1	651 x 60	
Base bridge y=100	1	651 x 60	
Base bridge y=170	1	651 x 60	
Base bridge y=280	1	651 x 60	
Base leg bridge y=-310	1	480 x 60	
Base leg bridge y=344	1	480 x 60	

5. Mark and drill

1. Print the drilling templates (`templates/base-templates.pdf`) at **actual size** and check the 100 mm square.
2. Tape each board's pages onto it: printed edges on the board's edges, grid lines lined up.
3. Awl through every crosshair. Pencil the dashed **pour line** on the walls.
4. Drill from the marked face, at the size the color legend gives. Pilot the big ones; the 63.5 mm holes are a 2-1/2 in hole saw.

The appendix has every board as a drawing and a table, if you would rather measure.

Hole	Drill	For
3.2 mm	1/8 in	motor-relief, split-gasket-groove
3.4 mm	9/64 in	corner-L, corner-R, gutter-L, gutter-R

Hole	Drill	For
4.5 mm	11/64 in	cleat, core post, corner bracket, corner-L, corner-R, draw-pocket-L, draw-pocket-R, gutter-L, gutter-R, leg holder, saddle, trough
5.5 mm	7/32 in	M5 insert screw, M5 screw into a rail bar
6.5 mm	17/64 in	M6 screw into a pad
9.6 mm	25/64 in	3/8 dowel, 3/8 dowel pin

6. Cut the foam

Fine saw, hot wire or a long knife. Stack and glue two layers of 2 in foam for the 90 mm cores. Wrap each piece smooth in packing tape. Cut a snug 12 mm square hole 30 mm deep for each core's post and glue the post in.

Casting	Piece	Qty	Size and how
base-front	center-a	1	XPS block 100 x 73 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	center-b	1	XPS block 100 x 72 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	center	1	XPS block 88 x 190 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	outboard-L	1	XPS block 120 x 174 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	trough-band-L	1	XPS block 101 x 55 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	draw-rod-L	1	XPS block 20 x 255 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	foot-b-L	1	XPS block 39 x 99 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	knee-L	1	XPS block 50 x 140 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	outer-a-L	1	XPS block 20 x 44 x 65 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	outer-b-L	1	XPS block 20 x 60 x 65 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	rear-L	1	XPS block 85 x 52 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	outboard-R	1	XPS block 120 x 174 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	trough-band-R	1	XPS block 101 x 55 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	draw-rod-R	1	XPS block 20 x 255 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.

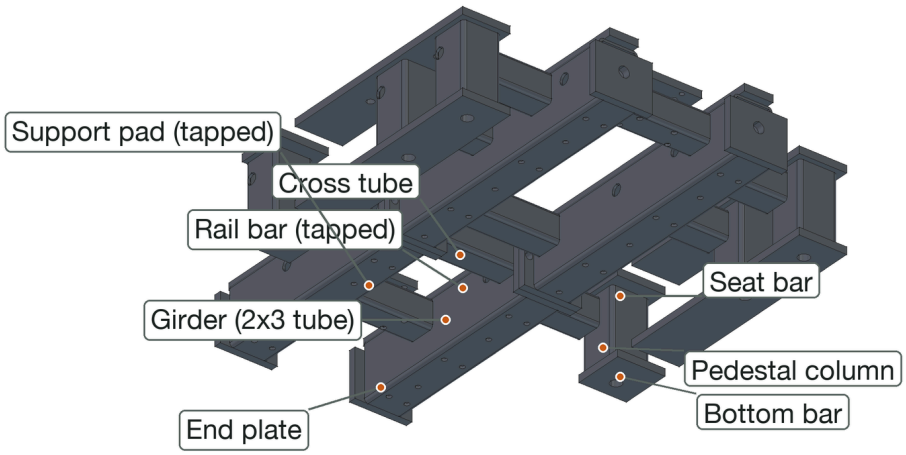
Casting	Piece	Qty	Size and how
base-rear	foot-b-R	1	XPS block 39 x 99 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	knee-R	1	XPS block 50 x 140 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	outer-a-R	1	XPS block 20 x 44 x 65 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	outer-b-R	1	XPS block 20 x 60 x 65 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	rear-R	1	XPS block 85 x 52 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
## 7. Wax			

- 3-5 coats of paste wax on every face the mix touches, boards and removable printed pieces, then PVA. Don't wax pieces that stay in (core posts, cast-in parts): scuff and degrease them.
- Glue quarter-round into the floor-to-wall corners that make outside edges; silicone fillets elsewhere.
- Grease every screw, bolt and rod that comes out of the casting.

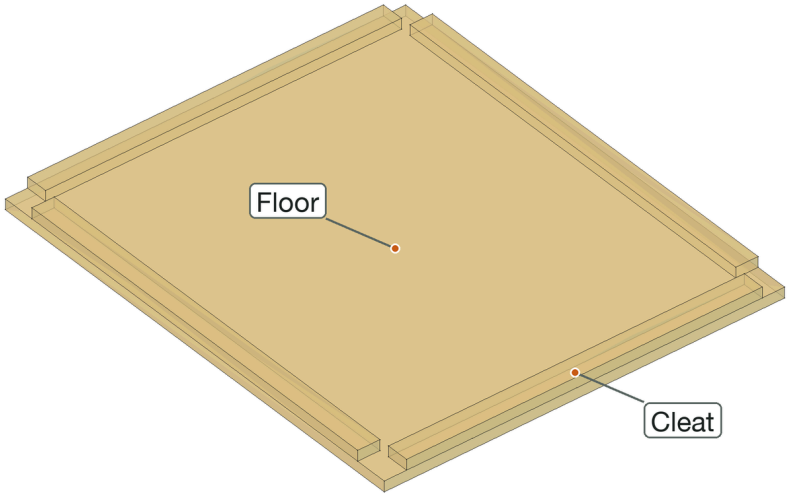
8. Assemble

Base, rear half (first)

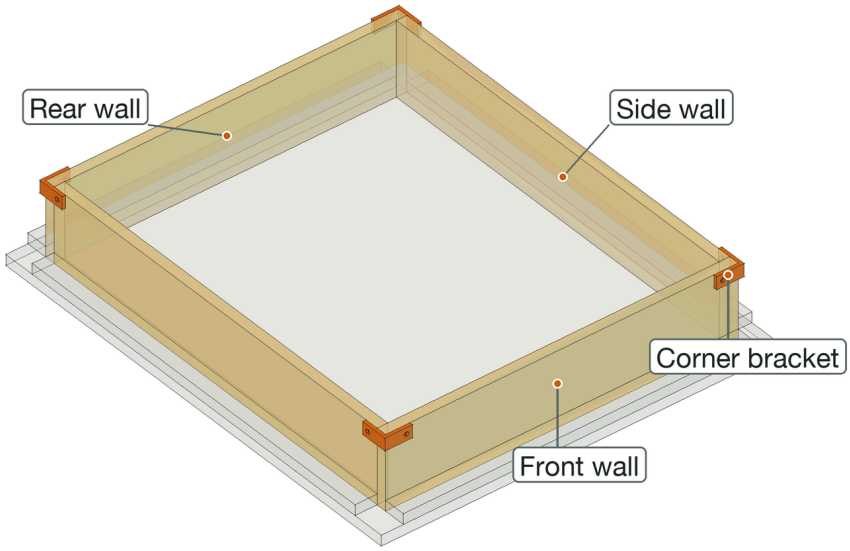
Cast upside down, top face on the floor; shown the right way up. New parts in color. Bolt each half's steel grid together first (the build guide, section 1), then it goes in as one piece.



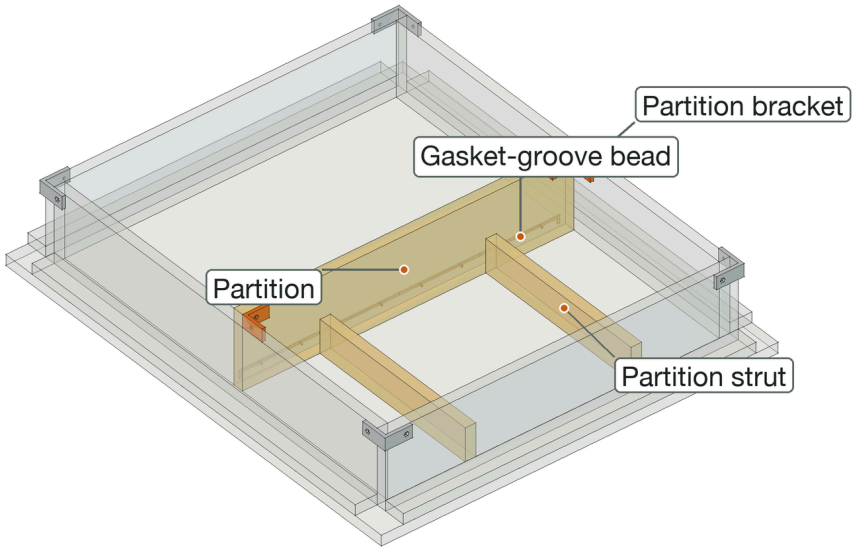
1. Floor on three battens. Bolt the cleats down.



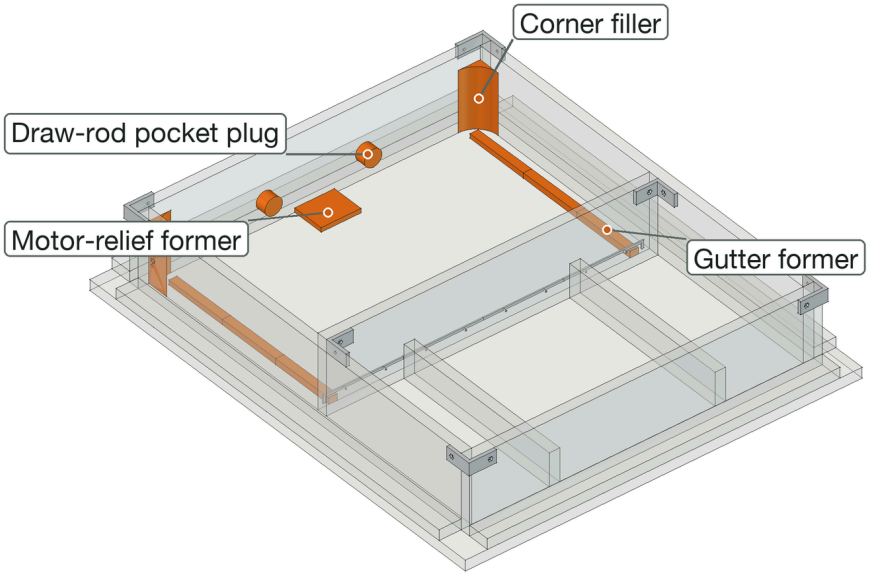
2. Walls against the cleats; corner brackets at the top. Diagonals equal within 2 mm, then tighten.



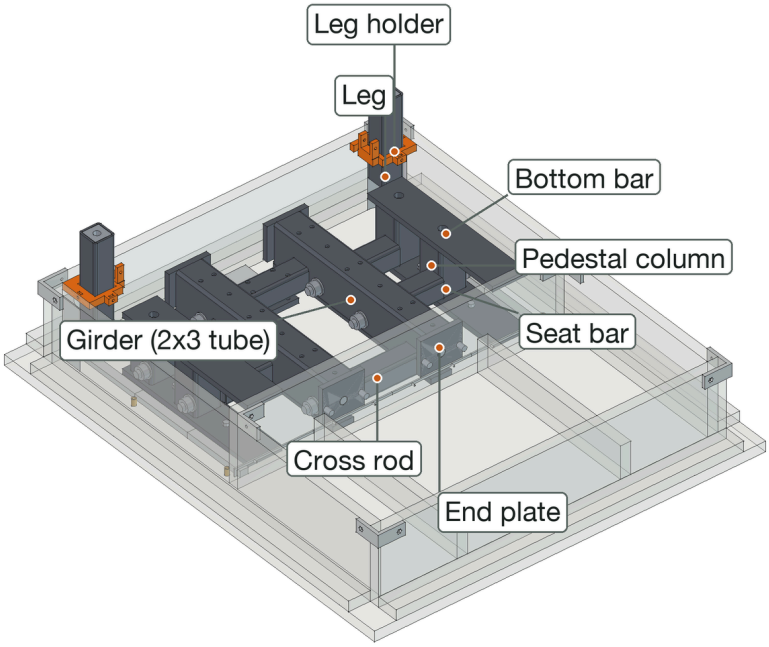
3. Partition between the side walls: two brackets, two struts to the front wall. Beads on its pins.



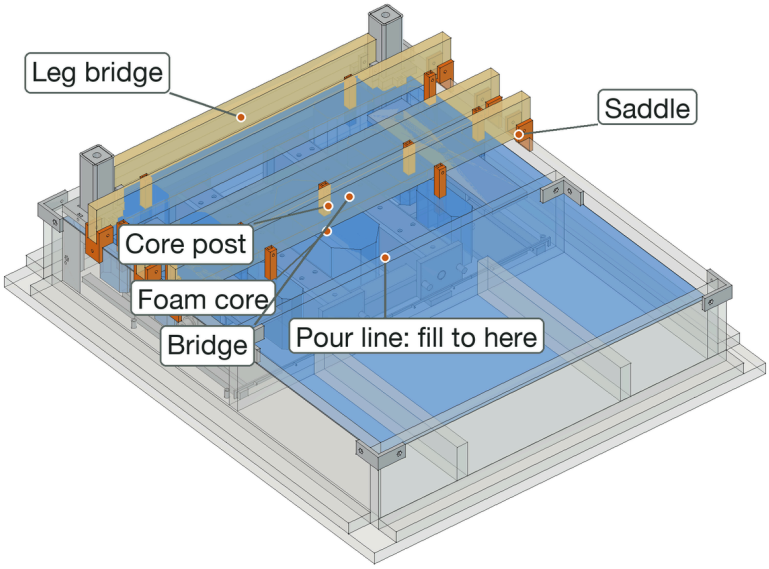
4. Gutter formers, motor relief, corner fillers and the draw-rod pocket plugs on.



5. The steel grid in, bolted up and upside down: its rail bars and the BK pad screwed to the floor (greased M5 and M6), its seats over the dowels, the split end plates against the partition with their dowel pins through it. Tape over the holes in the pedestals' bottom bars. Inserts; leg holders, legs set to height, pinch screws tight.

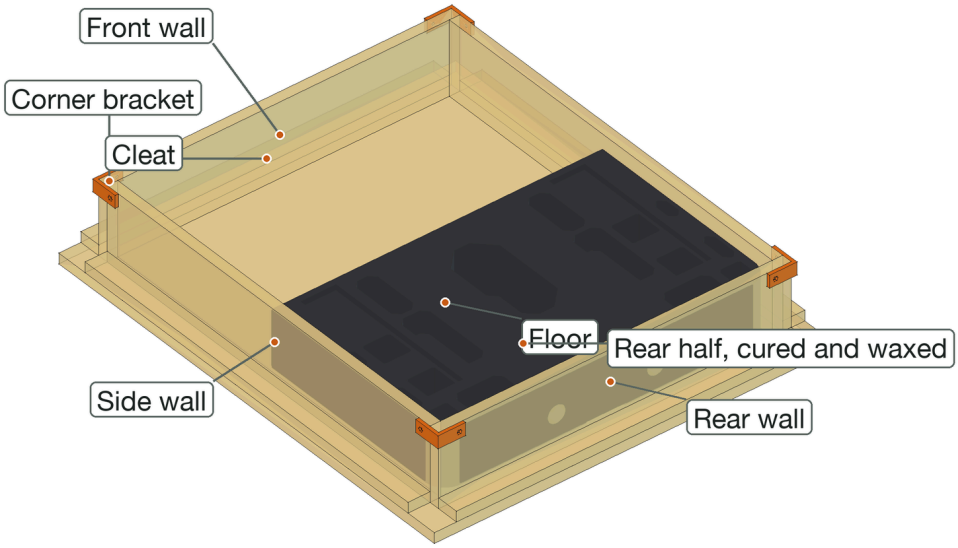


6. Pour the 20 mm skin. Foam cores on it between the steel, bridges into the saddles, each post bolted to its bridge. Fill to the pour line.

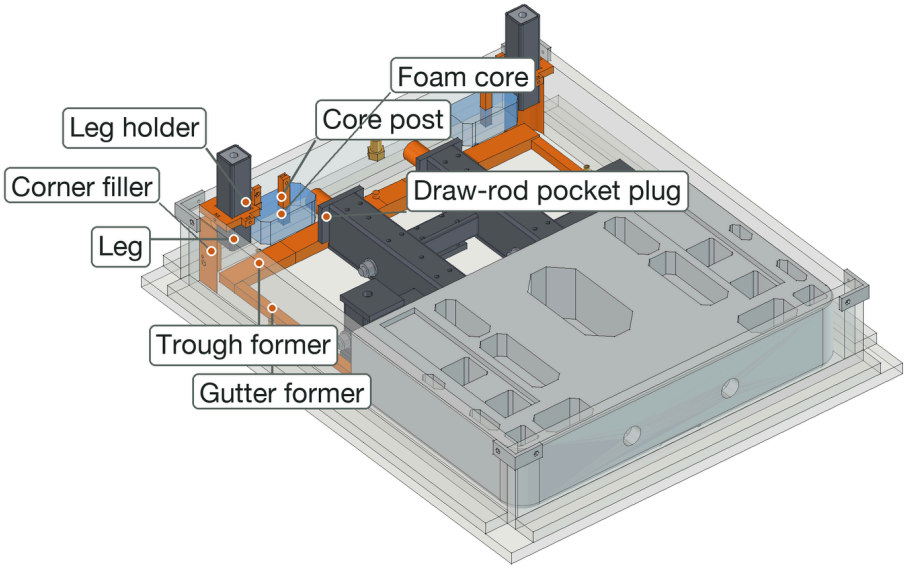


Base, front half (against the rear half)

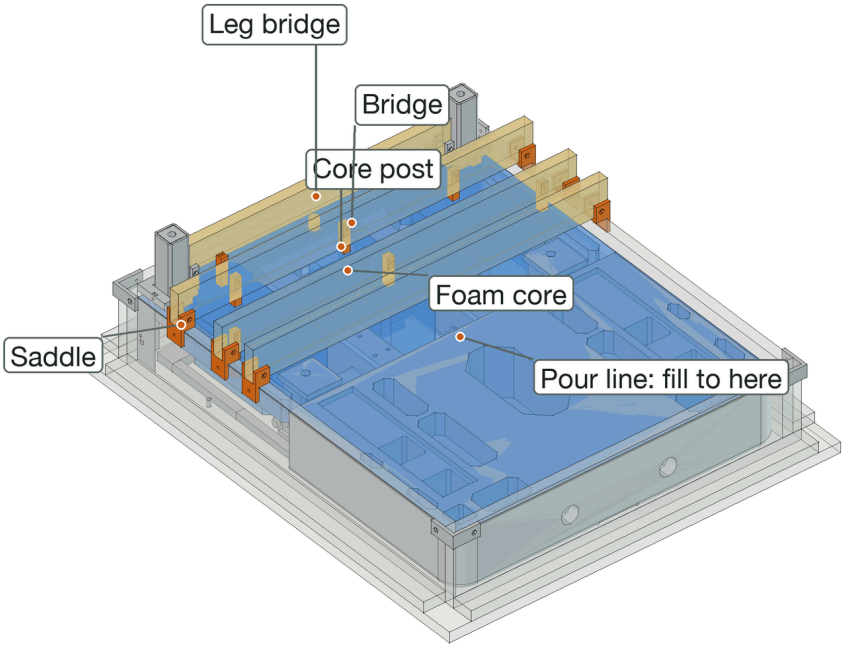
1. After the cure: bridges, partition and struts off. Pull the beads, tape the groove, and wax the rear half's split face, end plates and dowel pins.



2. Trough and gutter formers, corner fillers and pocket plugs on; the front grid in, its rail bars and BF pad screwed down and its split end plates onto the dowel pins; inserts and legs.

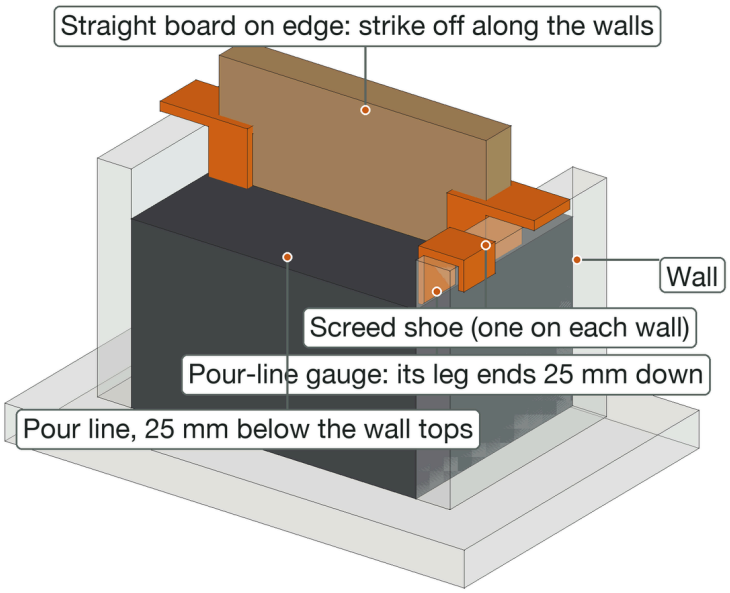


3. Skin, cores, bridges, fill. After the cure, take it apart and split the halves with wedges.



9. Pour

- Fill to the **pour line**, 25 mm below the wall tops (pencilled from the templates, or with the printed pour-line gauge clipped over a wall).
- Vibrate each layer until it stops bubbling; check the brackets halfway through.
- Strike the top off with a straight board in the two printed **screed shoes**: they hang it from the wall tops with its bottom edge on the pour line.



10. Take apart

1. After the full cure: bridges and saddles off, then corner brackets, then cleats; pull the walls off.
2. Screws, bolts and rods out from outside; lift the casting off the floor.
3. Unbolt and pull the formers and fillers.
4. Scrape and re-wax the boards; keep each mold's kit in a box. The kit is reused.

11. If something doesn't fit

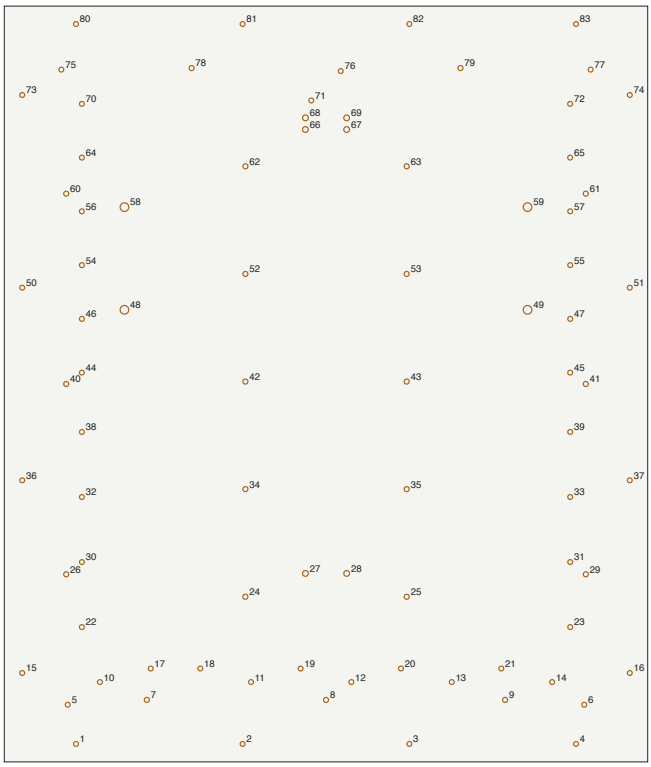
Problem	Fix
Nut won't slide under the hole	Clean the pocket with a nut on a bolt; warm the nut
Bracket or saddle loose on the board	Board thinner than set: tape shim, or set `mold_board` and reprint
Pin won't go in	Run the 1/8 in bit through again; trim the pin tip
Foam core moves	Tighten its post bolt; carpet tape under it

Problem	Fix
Printed part won't go on or into a steel tube	File the weld seam; sand the part. Several tight: caliper the tube, put its sizes in `frame_tube` / `post_tube`, re-run and reprint
Printed part loose on a tube	A wrap of tape round the part
Leak at a corner	Silicone fillet inside before waxing; check the brackets and diagonals
Former stuck after curing	Bolts out first, then a dowel and a light tap

Appendix: hole maps

Every board as seen from the face you mark: x from the left edge, y from the bottom edge (for a wall, the edge on the floor). The templates carry the same holes at full size.

Base floor



0,0 · 718 x 843 mm · seen from the cast side

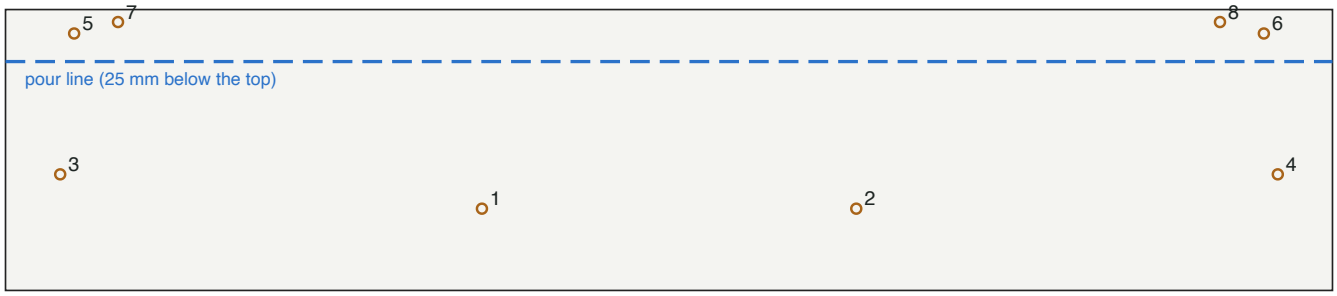
#	x	y	Hole	For
1	80	20	4.5	cleat
2	266	20	4.5	cleat
3	452	20	4.5	cleat
4	638	20	4.5	cleat
5	71	64	3.4	corner-R
6	647	64	3.4	corner-L
7	159	69	5.5	M5 insert screw

#	x	y	Hole	For
8	359	69	5.5	M5 insert screw
9	559	69	5.5	M5 insert screw
10	107	89	4.5	trough
11	275	89	4.5	trough
12	387	89	4.5	trough
13	499	89	4.5	trough
14	611	89	4.5	trough
15	20	99	4.5	cleat
16	698	99	4.5	cleat
17	163	104	4.5	trough
18	219	104	4.5	trough
19	331	104	4.5	trough
20	443	104	4.5	trough
21	555	104	4.5	trough
22	87	150	4.5	gutter-R
23	632	150	4.5	gutter-L
24	269	184	5.5	M5 screw into a rail bar
25	449	184	5.5	M5 screw into a rail bar
26	69	209	5.5	M5 insert screw
27	336	210	6.5	M6 screw into a pad
28	382	210	6.5	M6 screw into a pad
29	649	209	5.5	M5 insert screw
30	87	223	4.5	gutter-R
31	632	223	4.5	gutter-L
32	87	295	3.4	gutter-R
33	632	295	3.4	gutter-L
34	269	304	5.5	M5 screw into a rail bar
35	449	304	5.5	M5 screw into a rail bar
36	20	314	4.5	cleat
37	698	314	4.5	cleat
38	87	368	3.4	gutter-R
39	632	368	3.4	gutter-L
40	69	421	5.5	M5 insert screw
41	649	421	5.5	M5 insert screw
42	269	424	5.5	M5 screw into a rail bar

#	x	y	Hole	For
43	449	424	5.5	M5 screw into a rail bar
44	87	434	3.4	gutter-R
45	632	434	3.4	gutter-L
46	87	494	3.4	gutter-R
47	632	494	3.4	gutter-L
48	134	504	9.6	3/8 dowel
49	584	504	9.6	3/8 dowel
50	20	529	4.5	cleat
51	698	529	4.5	cleat
52	269	544	5.5	M5 screw into a rail bar
53	449	544	5.5	M5 screw into a rail bar
54	87	554	3.4	gutter-R
55	632	554	3.4	gutter-L
56	87	614	3.4	gutter-R
57	632	614	3.4	gutter-L
58	134	619	9.6	3/8 dowel
59	584	619	9.6	3/8 dowel
60	69	634	5.5	M5 insert screw
61	649	634	5.5	M5 insert screw
62	269	664	5.5	M5 screw into a rail bar
63	449	664	5.5	M5 screw into a rail bar
64	87	674	3.4	gutter-R
65	632	674	3.4	gutter-L
66	336	705	6.5	M6 screw into a pad
67	382	705	6.5	M6 screw into a pad
68	336	718	6.5	M6 screw into a pad
69	382	718	6.5	M6 screw into a pad
70	87	734	3.4	gutter-R
71	343	737	3.2	motor-relief
72	632	734	3.4	gutter-L
73	20	744	4.5	cleat
74	698	744	4.5	cleat
75	64	772	3.4	corner-R
76	375	770	3.2	motor-relief
77	654	772	3.4	corner-L

#	x	y	Hole	For
78	209	774	5.5	M5 insert screw
79	509	774	5.5	M5 insert screw
80	80	823	4.5	cleat
81	266	823	4.5	cleat
82	452	823	4.5	cleat
83	638	823	4.5	cleat

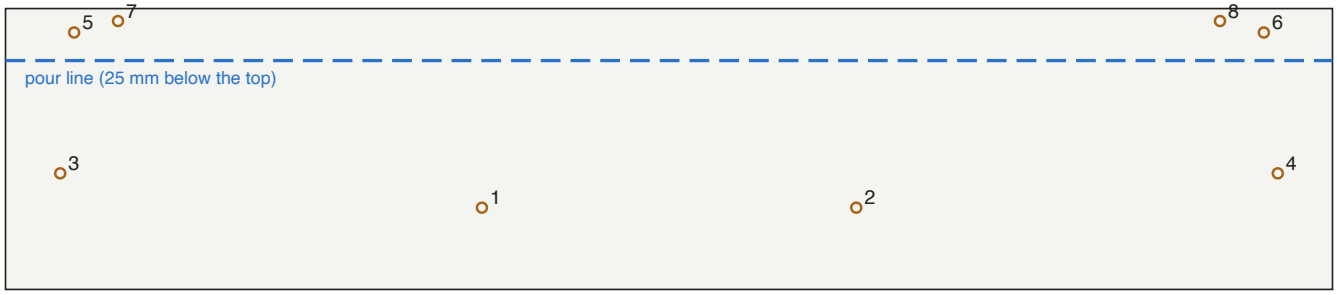
Base front wall



0,0 · 638 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	229	39	4.5	draw-pocket-L
2	409	39	4.5	draw-pocket-R
3	26	56	4.5	corner-L
4	612	56	4.5	corner-R
5	33	124	4.5	corner bracket
6	605	124	4.5	corner bracket
7	54	129	4.5	leg holder
8	584	129	4.5	leg holder

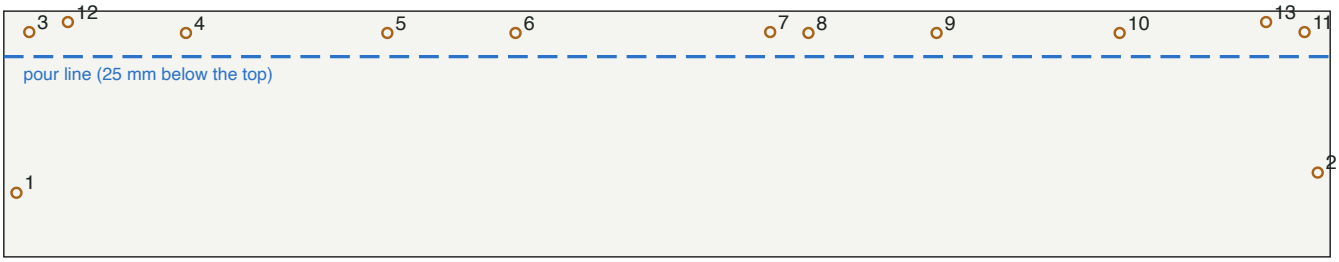
Base rear wall



0,0 · 638 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	229	39	4.5	draw-pocket-R
2	409	39	4.5	draw-pocket-L
3	26	56	4.5	corner-R
4	612	56	4.5	corner-L
5	33	124	4.5	corner bracket
6	605	124	4.5	corner bracket
7	54	129	4.5	leg holder
8	584	129	4.5	leg holder

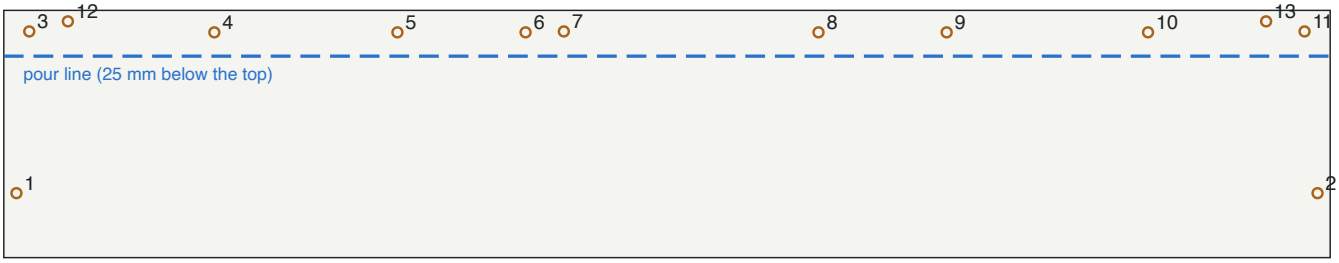
Base side wall, left



0,0 · 724 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	7	35	4.5	corner-L
2	718	46	4.5	corner-L
3	14	124	4.5	corner bracket
4	100	123	4.5	saddle
5	210	123	4.5	saddle
6	280	123	4.5	saddle
7	419	124	4.5	corner bracket
8	440	123	4.5	saddle
9	510	123	4.5	saddle
10	610	123	4.5	saddle
11	711	124	4.5	corner bracket
12	35	129	4.5	leg holder
13	690	129	4.5	leg holder

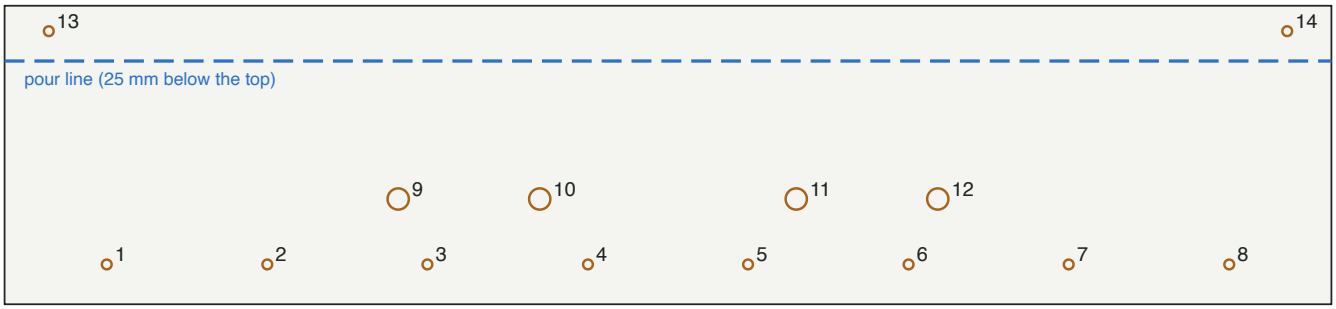
Base side wall, right



0,0 · 724 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	7	35	4.5	corner-R
2	718	35	4.5	corner-R
3	14	124	4.5	corner bracket
4	115	123	4.5	saddle
5	215	123	4.5	saddle
6	285	123	4.5	saddle
7	306	124	4.5	corner bracket
8	445	123	4.5	saddle
9	515	123	4.5	saddle
10	625	123	4.5	saddle
11	711	124	4.5	corner bracket
12	35	129	4.5	leg holder
13	690	129	4.5	leg holder

Base partition

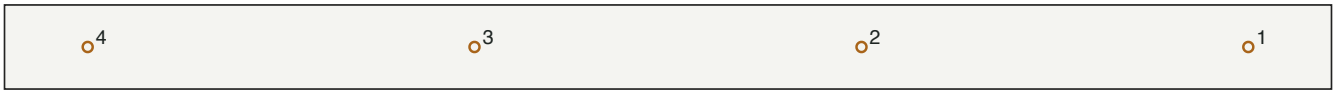


0,0 · 600 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	46	18	3.2	split-gasket-groove
2	119	18	3.2	split-gasket-groove
3	191	18	3.2	split-gasket-groove
4	264	18	3.2	split-gasket-groove

#	x	y	Hole	For
5	336	18	3.2	split-gasket-groove
6	409	18	3.2	split-gasket-groove
7	481	18	3.2	split-gasket-groove
8	554	18	3.2	split-gasket-groove
9	178	48	9.6	3/8 dowel pin (split)
10	242	48	9.6	3/8 dowel pin (split)
11	358	48	9.6	3/8 dowel pin (split)
12	422	48	9.6	3/8 dowel pin (split)
13	20	124	4.5	corner bracket
14	580	124	4.5	corner bracket

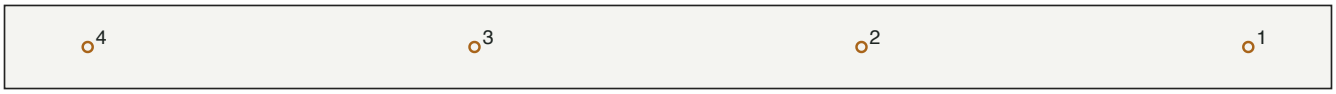
Base cleat, front



0,0 · 638 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	598	20	4.5	cleat
2	412	20	4.5	cleat
3	226	20	4.5	cleat
4	40	20	4.5	cleat

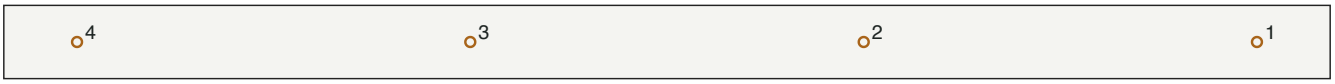
Base cleat, rear



0,0 · 638 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	598	20	4.5	cleat
2	412	20	4.5	cleat
3	226	20	4.5	cleat
4	40	20	4.5	cleat

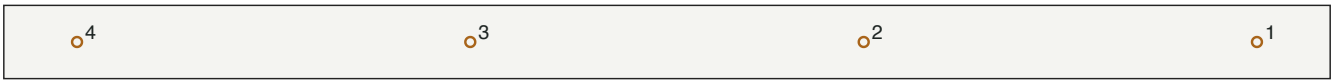
Base cleat, left



0,0 · 724 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	684	20	4.5	cleat
2	470	20	4.5	cleat
3	255	20	4.5	cleat
4	40	20	4.5	cleat

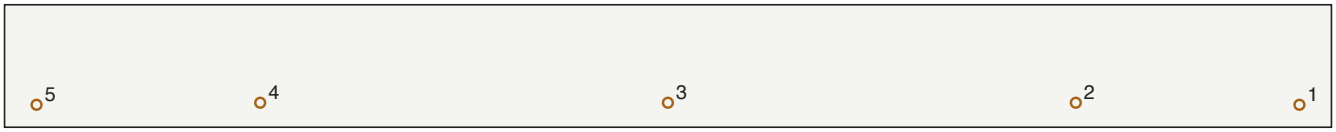
Base cleat, right



0,0 · 724 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	684	20	4.5	cleat
2	470	20	4.5	cleat
3	255	20	4.5	cleat
4	40	20	4.5	cleat

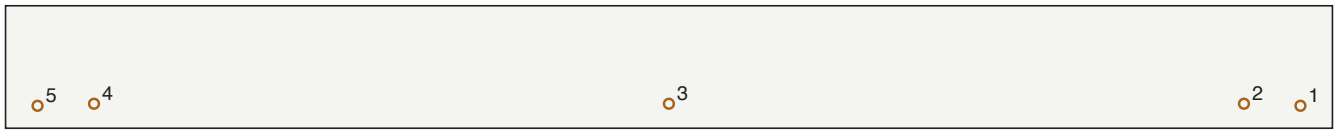
Base bridge y=-230



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	525	12	4.5	core post
3	325	12	4.5	core post
4	125	12	4.5	core post
5	16	11	4.5	saddle

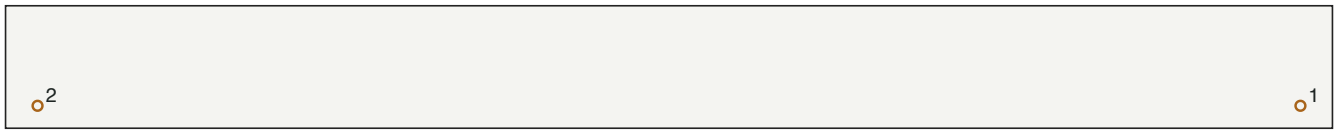
Base bridge y=-130



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	608	12	4.5	core post
3	325	12	4.5	core post
4	43	12	4.5	core post
5	16	11	4.5	saddle

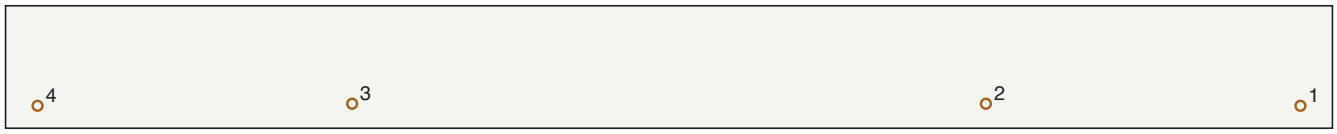
Base bridge y=-60



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	16	11	4.5	saddle

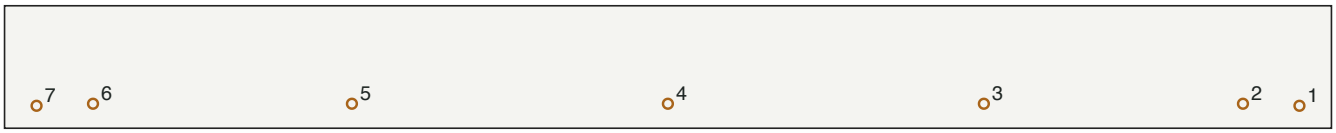
Base bridge y=100



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	481	12	4.5	core post
3	170	12	4.5	core post
4	16	11	4.5	saddle

Base bridge y=170



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	608	12	4.5	core post
3	480	12	4.5	core post
4	325	12	4.5	core post
5	170	12	4.5	core post
6	43	12	4.5	core post
7	16	11	4.5	saddle

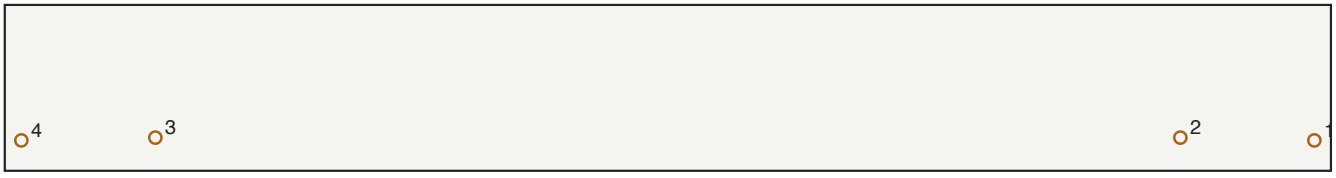
Base bridge y=280



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	608	12	4.5	core post
3	43	12	4.5	core post
4	16	11	4.5	saddle

Base leg bridge y=-310



0,0 · 480 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	474	11	4.5	leg holder
2	426	12	4.5	core post
3	54	12	4.5	core post
4	6	11	4.5	leg holder

Base leg bridge y=344



0,0 · 480 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	474	11	4.5	leg holder
2	428	12	4.5	core post
3	52	12	4.5	core post
4	6	11	4.5	leg holder